Title : Question Paper **FF No. 868**

Reg.No.						

Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: TY **Branch**: Computer

Subject: Design and Analysis of Algorithms **Subject Code:** CS 3205

Max. Marks:60 **Total Pages of Question Paper: 3**

Day & Date: Wednesday, 13/11/2024 **Time**: 2.30 pm to 4.30 pm

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Instructions to Candidate

1. Figures to the right indicate full marks.
2. Try to avoid using informal arguments. They won't get any credits. Try to write formal, concrete, and concise answers.
3. Use simple English language along with simple programming constructs while describing the algorithm. If the examiner cannot understand the programming constructs or the syntax you used, then naturally you will not get any credit. Your goal is to convey your algorithm to the examiner in the simplest possible language (of course without compromising concreteness, correctness)
4. While describing algorithms you must use only simple data structures like arrays, linked list, trees, etc. Any use of any advanced data structures like sets, hashmap, etc without giving implementation details carry no weightage.

Q. N.	CO No	BT* No		Max marks
Q. 1.				
Q.1. (a)	3	2, 3	For the following recurrences give best possible asymptotic upper bound on T(n) as a function of n. Express your answer using Big Oh asymptotic notation. (1) $T(n) = T(\sqrt{n}) + 1$ (2) $T(n) = 3T(n/2) + n$. (3) $T(n) = 4T(n/2) + n^2$	[6]
Q.1. (b)	3	2, 3	Give correct recurrence for f(n) for each of the following problems. Also find value of f(7) in both the cases. (1) A full binary tree is a binary tree in which every node has either zero or two children. Let f(n) denote the number of full binary trees on n nodes (nodes are not labeled). (2) Let f(n) be number of different ways to tile a grid of size 2 x n using rectangular tiles of size 1 x 2 (you can place 1 x 2 tiles vertically or horizontally).	[4]
Q. 2.				
Q. 2. (a)			Attempt Any **ONE** of the following.	
Q. 2. a)(i)	1, 2, 3	1, 3	Let $M_1, M_2, \ldots, M_k$ be the matrices such that dimension of matrix M_i is of dimension $n_i \times n_{i+1}$ for $1 \le i \le k$. There are several different ways to compute the matrix product $M_1.M_2 \ldots M_k$. Give an efficient algorithm to find out a best way (which uses minimum number of integer multiplications) to compute the product $M_1M_2 \ldots M_k$. Analyze time and the space complexity of your algorithm. Your algorithm shall output a root node of the binary tree T such that every internal node of the tree is a multiplication gate and leaf nodes are the matrices $M_1, M_2, \ldots, M_k$ such that the evaluation of the tree T corresponds to a best way to compute the product $M_1M_2 \ldots M_k$.	[6]

Q. N.	CO No	BT* No		Max marks
Q. 2. (a)(ii)	1,2,3	1,3	Consider the standard Tower of Hanoi game with n disks. There are 3 pegs A, B, C. There are n disks of distinct sizes placed in Peg A. The disks are numbered from 1 to n, where disk number n is the bottom most disk and disk number 1 is the topmost disk. There are the following two rules of the game. (1) In any move we can move topmost disk in any peg to other peg. (2) We cannot place a larger disk on top of a smaller disk. The goal of the game is to move all disks from Peg A to Peg C with minimum number of moves. Suppose you are given natural numbers n, k as input where $k \le 2^n - 1$. Give an efficient algorithm to find out which disk gets moved in the kth move in the standard solution of tower of Hanoi problem. Analyze time complexity of your algorithm. For full credit, time complexity of your algorithm (as a function of n) must be asymptotically equal to the time complexity of the best algorithm for the problem.		
Q. 2. (b)	5	1	Compare Las Vegas and Monte Carlo randomized algorithms. Give an example for each.	[4]
Q. 3.				
Q. 3. (a)			Attempt any **ONE** of the following.	[6]
Q. 3. (a) i)	1, 3, 6	3, 4, 5	What is the smallest value of p<1 for which the following statement regarding Huffman encoding scheme is TRUE? If some character occurs with frequency more than p, then there is guaranteed to be a codeword of length 1. Prove the above statement for the claimed value of p.	
Q. 3. (a) ii)	2,3,6	3,4	Consider a 2-D array (grid) of size n x n containing distinct integers. Describe an algorithm to find a sub-rectangle of the grid with the largest possible sum (sub-rectangle contains contiguous rows and columns). Analyze the time complexity of your algorithm. Note that for full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	
Q. 3. (b)	1,2, 3	1, 3	Consider Josephus game with n players. Players are numbered from 1 to n. Players are standing on a circle with first player holding a sword. Player 1 kills player 2 and passes the sword to player 3. Player 3 kills player 4 and passes sword to player 5. this goes on, any player having the sword kills next alive person in clockwise direction and passes the sword to the next alive person in clockwise direction. The game continues till only one player is left, and this player is called as the winner of the game. Let f(n) denotes winner of Josephus game with n players. (1) Given natural number n, briefly describe an efficient algorithm for finding f(n) given n as input. (2) Who is winner of the Josephus game with $n = 2^{700} + 2^{100} - 1$ players?	[4]
Q. 4.				
Q.4. (a)	1, 4, 6	3, 4	Suppose there exist a language L in NP such that (1) L is not equal Σ^* (2) L is not equal to empty set (3) L is not NP-complete then prove that P is not equal to NP.	[6]
Q. 4. (b)	1,4	1, 2	Give Prover Verifier based definition for the NP class and give two examples of the problems characterizing the NP class. For each of the problems clearly describe what is the input and the expected output of the problem.	[4]

Q. N.	CO No	BT* No		Max marks
Q. 5.				
Q.5. (a)	1, 3	1,3	Define greatest common divisor (gcd) of two natural numbers m, n. Prove that gcd of any successive terms in the Fibonacci series is 1.	[3]
Q.5. (b)	2,3	1,3	Briefly describe the fast modular exponentiation algorithm. Find a remainder one obtains on dividing 4444^{4444} by 9.	[3]
Q.5. (c)	1,2	1, 3	What is the Optimal Binary Search Tree (OBST) construction? Clearly describe the problem and give a concrete recurrence relation for the optimal search cost in a BST.	[4]
Q. 6.				
Q. 6. (a)	1, 2, 3, 4, 6	3, 4	Give an algorithm to print all the subsets of the given set {A[1], A[2], …, A[n]}. Give proper pseudocode for the algorithm. How may subsets a set of cardinality n would have? Prove your claim. Analyze time and space complexity of your algorithm.	[6]
Q. 6. (b)	1, 3, 5	3, 4	Consider a planet X where there are n days in a year and every person on the planet has a birthday uniformly distributed over these n days. Suppose among any p people on the planet X, with probability at least 1/2 there are at least two people with the same birthday. Show that the smallest possible value for p is $\Theta(\sqrt{n})$ where Θ denotes the Big Theta Asymptotic notation.	[4]

CO Statements:

1. To formulate computational problems in abstract and mathematically precise manner
2. To design efficient algorithms for computational problems using appropriate algorithmic paradigm
3. To analyze asymptotic complexity of the algorithm for a complex computational problem using suitable mathematical techniques.
4. To establish NP-completeness of some decision problems, grasp the significance of the notion of NP-completeness and its relationship with intractability of the decision problems.
5. To understand significance of randomness, approximability in computation and design randomized algorithms for simple computational problems and design efficient approximation algorithms for standard NP-optimization problems.
6. To incorporate appropriate data structures, algorithmic paradigms to craft innovative scientific solutions for complex computing problems.

***Blooms Taxonomy (BT) Level No:**

1. Remembering; 2. Understanding; 3. Applying; 4. Analyzing; 5. Evaluating; 6. Creating